SCHOOL OF ENGINEERING AND BUILT ENVIRONM

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COMPUTER PROGRAMMING SEMESTER PROJECT

System Requirements specification (SRS)

SAFETY HAZARD AWERENESS AND PREVENTION APP (SAFE SPHERE)

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1.Introduction

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Purpose of the document

This document defines the software requirements for the Safety Hazard Awareness and Prevention App. It outlines the system's functionality, performance expectations, and constraints to guide development and ensure all stakeholders understand the system.

Scope

The app aims to:

- Educate users about safety hazards in mines and industrial environments
- Provide safety training courses
- Track user progress in safety learning
- Help prevent accidents by preparing users before entering hazardous work sites

The app does not:

- Replace official workplace certification
- Provide real time hazard detection on site
- Act as an emergency response system

Engineering domain and problem statement

Domain: Industrial safety

Problem statement:

Many accidents in mines and industrial workplaces occur due to lack of safety awareness and proper training. Workers and visitors are often exposed to hazards without adequate preparation. This app aims to reduce such incidents by providing accessible safety education, training courses and progress tracking before entering hazardous environments.

2.System Overview

System description

Safe Sphere is a mobile safety awareness and prevention application designed to reduce accidents in mining and industrial work environments. The system is built around the principle of "prevention instead of reaction," aiming to educate users about potential hazards before they enter high-risk areas.

It is intended to support students, workers, and visitors by equipping them with the knowledge required to safely navigate hazardous sites.

Users can sign in, access safety content, complete courses and track their progress. The app stores data using cloud services and allows users to learn anytime before visiting a work site.

Target users

- Mine workers
- Industrial workers
- Engineering students
- Site visitors

Technical level

- Basic smartphone users
- No advanced technical skills required

Environment

- Android smartphones

- Internet access required

Technology overview

- Frontend: Expo + React Native
- Backend: Firebase
- Database: Fire store
- Authentication: Firebase Authentication
- Storage: Firebase storage

- **Firebase Data Model Collections:**

- Users: user ID, name, email, progress level
- Courses: course ID, title, description, content
- Progress: user ID, course ID, completion status, score
- Reports (optional): report ID, user ID, hazard type,

description

Assumptions and Constraints Assumptions:

- Users have smartphones
- Users can read and understand English
- Internet access is available

Constraints

- Requires android 10+
- Requires internet for syncing data
- Limited offline functionality

3.Functional Requirements

ID	Requirement statement	Priority	Firebase service

FR-001	The system shall allow users to register using email password	high	Firebase auth
FR-002	The system shall allow users to login and log out securely	high	Firebase auth
FR-003	The system shall allow users to view available safety course	high	Firestore
FR-004	The system shall allow users to access course contents (text /video)	high	Firestore
FR-005	The system shall track user progress for each course	high	Firestore
FR-006	The system shall display user progress and completion status	high	Firestore

FR-007	The system shall allow users to take quizzes after completing lessons	high	Firestore
FR-008	The system shall store quizzes results and scores	medium	Firestore
FR-009	The system shall allow users to view safety related images/ videos	medium	Firebase storage
FR-010	The system shall provide safety	high	Firestore
	tips and hazard information		

4.Non – Functional Requirements

Performance

- This app shall load the main dashboard within 5 seconds on a stable internet connection
- The app shall allow users to navigate between screens without noticeable delays
- The system shall support at least 1000 concurrent users without performance degradation

- Data retrieval from the database shall occur within 2 to 3 seconds

Security

- All user data shall be protected using Firebase Authentication
- Only authenticated users can access content
- The system shall prevent unauthorised access to user accounts
- Sensitive data e.g. passwords shall not be stored in plain text
- The system shall automatically log out users after a period of inactivity

Usability

- The app shall have a simple intuitive interface
- A new user should navigate the app without training
- The app shall provide clear instructions for all features
- Icons and labels shall be easily understandable
- The system shall support basic accessibility (readable fonts, clear contrast)
- Error messages shall be cleared and guide users on how to fix issues

Compatibility

- The app shall run on android 10 and above
- The app shall be developed using React Native, allowing cross platform compatibility
- The system shall support IoS devices (iPhones) in future versions

- The app shall function across different screen sizes and resolutions
- The system shall work on various smartphone brands

Reliability

- User progress data shall not be lost during temporary network issues
- The system shall recover gracefully from errors
- Data shall be automatically saved when users complete lessons or quizzes
- The system shall maintain data integrity at all times

Maintainability

- The code shall be well structured and documented
- Updates shall be easy to implement
- New features shall be added without affecting existing features
- The system shall follow modular design principles

Availability

- The app shall be available 24/7, except during maintenance
- Downtime shall be minimised
- The system shall notify users during maintenance

Portability

- The app shall be installed on different Android devices

Safety

- The system shall provide accurate and reliable safety information

- Content shall be regularly updated to reflect current safety standards
- The app shall not provide misleading or incorrect safety guidance
- The system shall clearly indicate that it does not replace

certified safety training

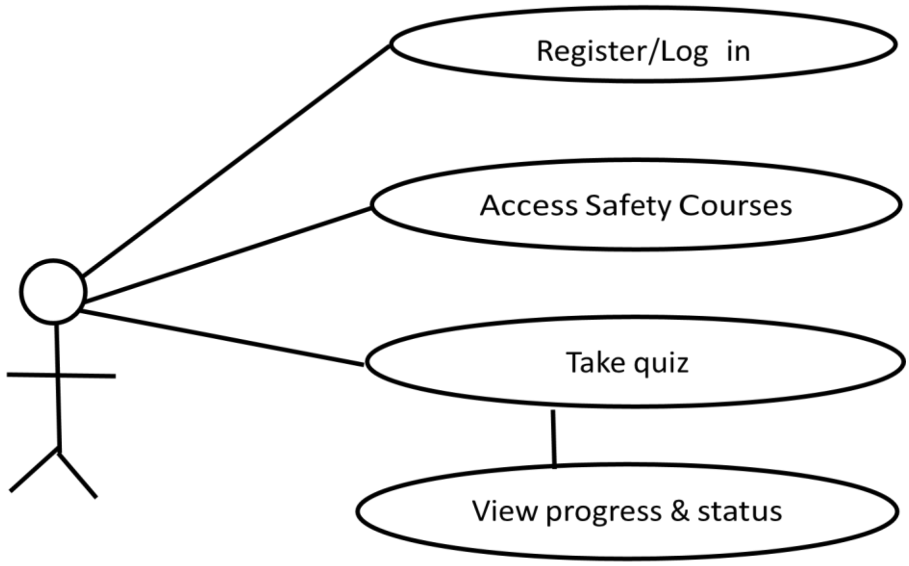
Network

- The app shall require an internet connection for most features
- Some content may be cached for limited offline access

5.Use Case Diagram

Primary actors

- User (worker, student, or visitor)



Top 3 use cases 1. Register/ Log in

This use case allows a user to create an account or log into the system to access safety courses and track progress

- User creates an account or logs in

- System authenticates user
- User gains access to the app

2. Access Safety Courses

This use case allows users to browse and access safety training content related to mining and industrial environments

- User selects a course
- System displays course content
- User studies material

3. Track Progress

This use case allows users to complete the quizzes after lessons and track their progress within the app

- User completes lessons/ quizzes
- System records progress
- User views progress status

Conclusion

Safe Sphere is designed to enhance safety awareness and reduce accidents in mining and industrial environments by focusing on prevention rather than reaction. The application provides users with access to safety information, training courses, and progress tracking to ensure they are well-prepared before entering high-risk areas.

By integrating user-friendly features such as account management, learning modules, and cloud-based data storage, the system supports flexible and accessible learning for students, workers, and visitors.

Overall, Safe Sphere aims to promote a safer working environment by equipping users with the necessary knowledge and tools to identify and avoid potential hazards, thereby contributing to improved safety standards in industrial settings.